UNIT 4 — CLOUD SECURITY

Practical Report: Launch a Windows VM and Connect Using RDP

Purpose: provision a Windows Server EC2 instance, connect safely over RDP, verify basic security settings, and document the work.

Important screenshot note

The interface images in this report are realistic-style **illustrative mockups**, not captures from a live AWS account or an actual Windows VM. Replace them with your own screenshots before submitting if your instructor requires evidence of your completed lab.

Prepared as a step-by-step practical guide

1. Objective

Launch a Windows Server virtual machine on Amazon EC2, connect over Remote Desktop Protocol (RDP), inspect system information, create a local user, verify Windows Firewall, and create a test file.

2. Launch a Windows Server EC2 instance

• Open AWS Console → EC2 in the intended region and choose Launch instance.

• Select the Windows Server AMI required by your instructor (for example, Windows Server 2022 Base), then choose a lab-appropriate instance type.

• Create or select a key pair. AWS uses the private key to decrypt the initial Administrator password; keep the key private.

• Configure a security group with inbound RDP TCP 3389 sourced only from My IP (/32). Do not use 0.0.0.0/0 for RDP.

• Launch the instance and wait for the instance status checks to pass. Record the public IP/DNS and instance ID for your own report.

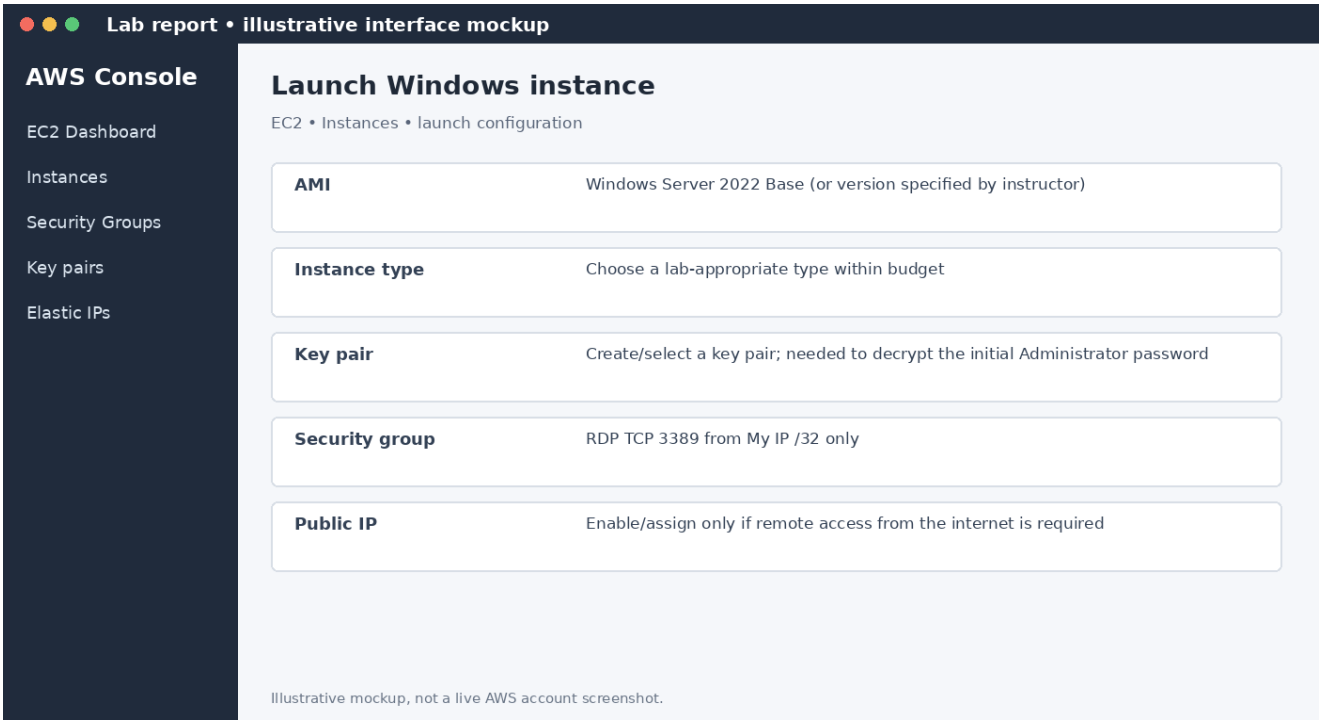


Figure 1. Example Windows EC2 launch configuration. Confirm exact settings in your own AWS account.

3. Configure RDP access securely

Open EC2 → Security Groups → Inbound rules. Confirm TCP port 3389 has your current public IP as the source. If your IP changes, update the rule. A VPN or managed access method is preferable where available.

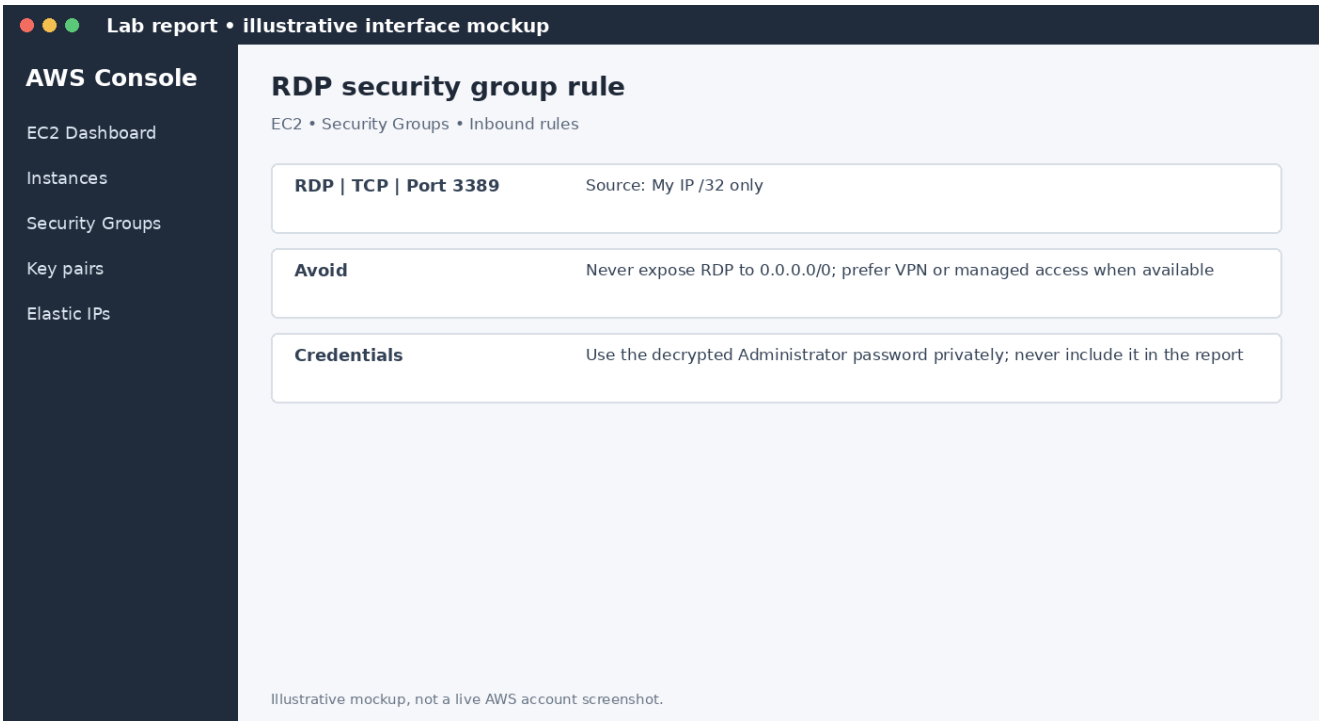


Figure 2. Example RDP security group. The IP source must be your real public IP with /32.

4. Retrieve credentials and connect with Remote Desktop

• Select the running instance and choose Connect → RDP client (console wording may vary).

• Use Get password / Decrypt password and your private key to retrieve the initial Administrator password. Never put this password in the report or screenshots.

• On Windows, open Remote Desktop Connection (mstsc), enter the instance public IPv4 address or DNS name, and connect. Sign in as Administrator with the decrypted password.

If RDP fails, verify the instance status checks, public address, route to the internet, source IP in the security group, and TCP 3389 reachability. Do not weaken the rule to all internet addresses.

5. Explore the desktop and check system information

On the remote desktop, open Server Manager or run the following in PowerShell. Record the actual values returned by your VM rather than copying example values.

```
Get-ComputerInfo | Select-Object WindowsProductName, WindowsVersion, CsName
systeminfo
```

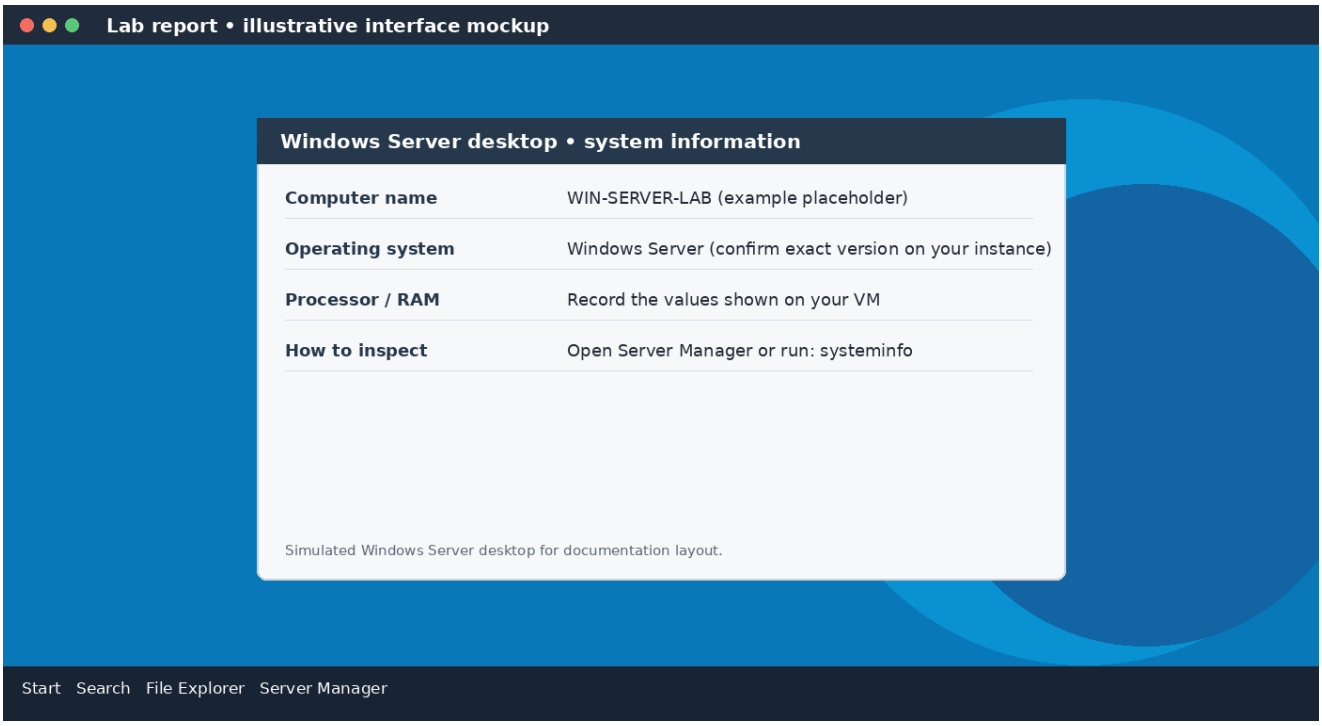


Figure 3. Illustrative Windows Server desktop and system-information layout. Replace with a screenshot from your connected VM.

6. Create a local user account

Open PowerShell as Administrator and run the command below. Enter a unique strong password when prompted; do not include it in the report. Follow your institution's password policy.

```
$Password = Read-Host -AsSecureString 'Enter a strong password'
New-LocalUser -Name 'labuser' -Password $Password -FullName 'Lab User' -Description
'Unit 4 practical user'
Get-LocalUser -Name 'labuser'
```

Use a different username if required by your lab. Do not add the user to Administrators unless the assignment explicitly requires it.

7. Enable and verify Windows Firewall

```
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True
Get-NetFirewallProfile | Select-Object Name, Enabled
```

The Enabled column should show True for the profiles you enabled. Make sure firewall configuration does not accidentally block required lab access.

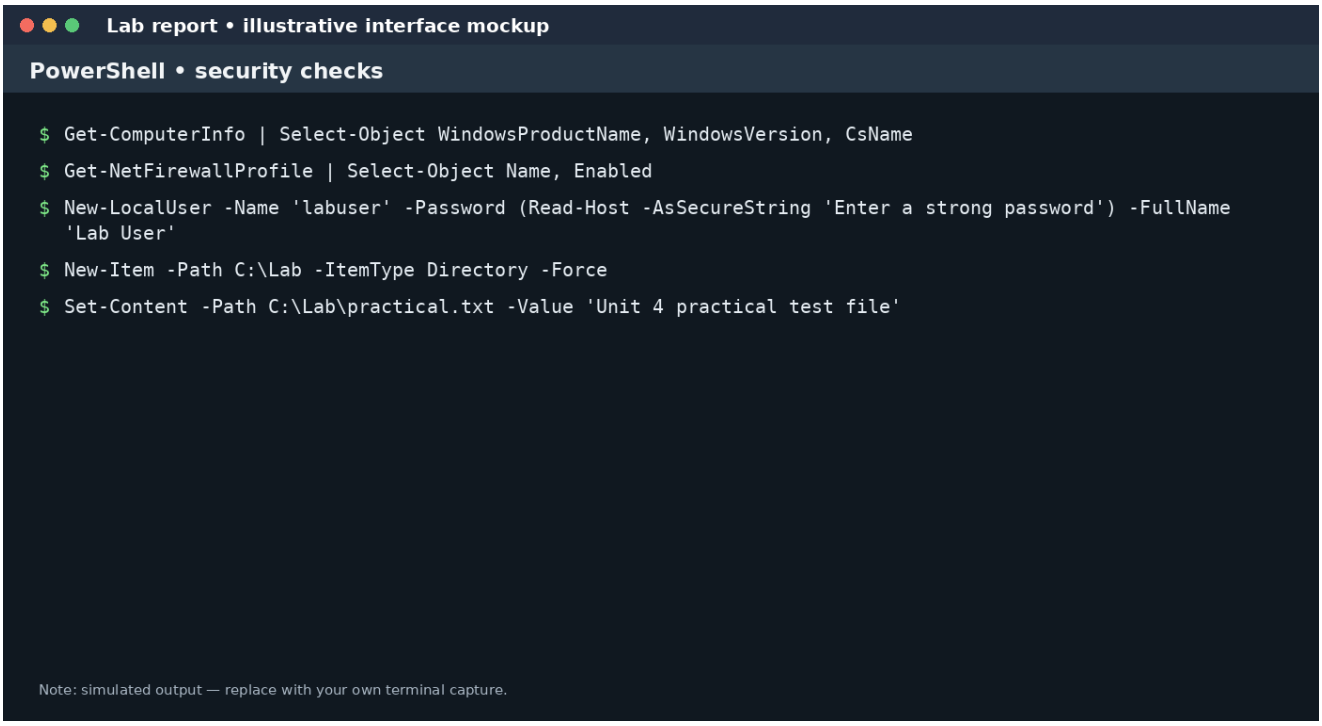


Figure 4. Example PowerShell commands for system checks, firewall verification, user creation, and a test file. Simulated output; capture your own results.

8. Create a test file

```
New-Item -Path C:\Lab -ItemType Directory -Force
Set-Content -Path C:\Lab\practical.txt -Value 'Unit 4 practical test file'
Get-Content C:\Lab\practical.txt
```

Open File Explorer and navigate to C:\Lab to show the created file if a graphical screenshot is required.

9. Screenshot checklist

Capture your own evidence for: (1) EC2 instance details, (2) inbound RDP rule showing your IP /32, (3) the connected Windows desktop/system information, (4) PowerShell output showing the local user and firewall state, and (5) the created test file. Do not include the Administrator password, private key, or other credentials.

10. Cleanup and conclusion

When finished, stop or terminate the VM to avoid unnecessary charges. Stopping may preserve some storage charges; terminating is irreversible. The practical is complete when you can connect over restricted RDP, inspect system information, create the local account, confirm Windows Firewall is enabled, and create/read the test file.